

Codefest 8

1. Aggregate spike rate = $N \times f = 1024 \times 50 = 51200$ spikes/second
2. Mean required AER = Aggregate spike rate $\times 20 = 51200 \times 20 = 1024000$ bit/s
 $= 1.024 \text{ Mbit/s}$

3. All the three interfaces can sustain the mean rate of 1.024 Mbit/s .
 I^2C is the lowest complex interface and its 3.4 Mbit/s is more than ideal for the mean rate of 1.024 Mbit/s

4. Peak bandwidth = $\frac{0.25 \times 1024 \times 20}{0.001} = 5.125 \text{ Mbit/s}$

The peak bandwidth is higher than the mean required AER bandwidth.

$$\text{Ratio} = \frac{\text{Peak bandwidth}}{\text{Mean AER bandwidth}} = \frac{5.125}{1.024} \approx 5$$

The I^2C maximum speed mode is 3.4 Mbit/s which is less than 5.125 Mbit/s which means a buffer is required.

The buffer depth is $5.125 - 3.4 = 1.725 \text{ Mbit/s}$.

5. Frame bandwidth = $\frac{1024 \times 1 \text{ bit}}{0.001} = 1.024 \text{ Mbit/s}$

$$\text{Ratio} = \frac{\text{Frame bandwidth}}{\text{AER bandwidth}} = \frac{1.024 \text{ Mbit/s}}{1.024 \text{ Mbit/s}} = 1$$

$$N \times f \times 20 = N \times 1000 \Rightarrow f = \frac{1000}{20} = 50$$

When AER is the right choice the bandwidth scales when there is activity rather than staying fixed.